Lecture #10: Inequalities, Convergence and the Central Limit Theorem

MATH 321 – Mathematical Statistics I

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Inequalities

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If we know a probability distribution (either the PDF or the CDF), we know, in theory, all the probabilities associated with it.

But what if we only know some summary statistics for X?

We will use some theoretical inequality results to develop this idea.

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Markov's Inequality

Theorem: Let $X \geq 0$ be a non-negative random variable. Then, for any $\varepsilon > 0$, we have

$$P[X \geq \varepsilon] \leq \frac{E[X]}{\varepsilon}.$$

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Example: The number of items sold in a week is a random variable with mean 16. Estimate the probability that next week's sales exceed 25.

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Chebyshev's Inequality

Theorem: Let X be a random variable with mean μ and finite variance $\sigma^2 = V[X]$. Then for any $\varepsilon > 0$, we have

$$P\big[|X - \mu| \geq \varepsilon\big] \leq \frac{V[X]}{\varepsilon^2}.$$

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If we set $\varepsilon = k\sigma$ for some $k > 0$, we get the more familiar form of Chebyshev's Inequality:

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Example: Let $X \sim \text{Uniform}(0,\, 2\sqrt{2})$. Find the bound of Chebyshev's inequality for the probability $P\big[|X - \mu| \geq 1\big]$.

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Chernoff's Bound

Theorem: Let X be a random variable whose moment-generating function $M_X(s)$ exists. Then, for any $\varepsilon \geq 0$, we have that

$$P[X \geq \varepsilon] \leq e^{-\Lambda^*(\varepsilon)},$$

where

$$\Lambda^*(\varepsilon) = \sup_{s>0} \Big\{ s\varepsilon - \log M_X(s) \Big\}.$$

$\Lambda^*(\varepsilon)$ is called the Fenchel–Legendre dual function of $\log M_X$.

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Hoeffding's Inequality

Theorem: Let $X_1, X_2, \ldots, X_N$ be i.i.d. random variables with $0 \leq X_i \leq 1$ and $E[X_i] = \mu$. Then, for any $\varepsilon > 0$,

$$P\left[|\overline{X}_N - \mu| > \varepsilon\right] \leq 2e^{-2\varepsilon^2 N},$$

where $\overline{X}_N = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} X_i$.

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Sample Averages

The **sample average** of a sequence of random variables $X_1, \ldots, X_N$ is

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This is a linear combination of random variables, so it too is a random variable.

Thus, it has an expectation, variance, PDF, CDF and everything.

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Sample Averages

If $X_1, \ldots, X_N$ are i.i.d., then the summary statistics $E(\overline{X}_N)$ and $V(\overline{X}_N)$ are easy.

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If $X_1, \ldots, X_N$ are i.i.d., then the summary statistics $E(\overline{X}_N)$ and $V(\overline{X}_N)$ are easy.

But what is the ***probability distribution*** of $\overline{X}_N$?

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The Weak Law of Large Numbers

Theorem: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with mean μ and finite variance σ^2. Let $\overline{X}_N = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} X_i$. Then, for any $\varepsilon > 0$,

$$\lim_{N\to\infty} P\Big[|\overline{X}_N - \mu| > \varepsilon\Big] = 0.$$

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Convergence in Probability

A sequence of random variables $A_1, A_2, \ldots$ **converges in probability** to a deterministic number α if for every $\varepsilon > 0$,

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Example: Let $X_1, X_2, \ldots$ be i.i.d. random variables with $X_i \sim \text{Uniform}(0, 1)$. Define $A_N = \min\Big\{X_1, \ldots, X_N\Big\}$. Show that A_N converges in probability to zero.

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We write $A_N \xrightarrow{a.s.} \alpha$ to denote almost sure convergence.

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The Strong Law of Large Numbers

Theorem: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with common mean μ and variance σ^2. Assume $E[X_i^4] < \infty$. Let $\overline{X}_N = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} X_i$ be the sample average. Then

$$P\Big[\lim_{N\to\infty} \overline{X}_N = \mu\Big] = 1.$$

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The Central Limit Theorem

Theorem: Let $X_1, X_2, \ldots$ be a sequence of i.i.d. random variables with mean $E[X_i] = \mu$ and variance $V[X_i] = \sigma^2 > 0$. Also assume that $E\left[|X_i|^3\right] < \infty$. Let $\overline{X}_N = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} X_i$ be the sample average and let $Z_N = \sqrt{N}\left(\dfrac{\overline{X}_N - \mu}{\sigma}\right)$. Then, for every $z \in \mathbb{R}$,

$$\lim_{N\to\infty} F_{Z_N}(z) = F_Z(z),$$

where $Z \sim \text{Normal}(0, 1)$.

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The Central Limit Theorem Examples

Examples:

1. Suppose we have $X_1, \ldots, X_{30} \overset{i.i.d.}{\sim} \text{Binomial}(n = 20, p = 0.25)$. What is the probability that the sample mean of these 30 variables is above 8?

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2. American males are, on average, 69 inches tall, with a standard deviation of 3 inches. If I select a sample of 9 American males, what is the probability that our sample mean is within 0.5 of an inch of the population mean?

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3. Suppose I want my sample mean to be within 1 inch of the true population mean, with 95% probability. How many people do I need to include in my sample?

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